Writing scientific papers

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Outline

Scientific work

Publication

SEMCON

Structuring a scientific paper

Summary - timeline

A brief history of electromagnetism

The good old days

600 BC

THALES OF MILETUS
He was a philosopher also known as one of the legendary seven wise men, of antiquity, who rubbed the amber and the fur and the bits of fur and hair attracted.

AFTER 600 BC

MAGNES THE SHEPERD
Magnes was a shepered that discovered magnets while taking care of his sheep. And that also discovered lodestones that contain magnetites that where natural.

1600

WILLIAM GILBERT
Also known as gilberd, William was an english physician, physicist, and natural philosopher who published "The Magnete" and described the earth as a magnet.

1747

BEN FRANKLIN (1706-1790)
From 1706-1790 Benjamin Franklin was a statesman, author, publisher, scientist, inventor, and diplomat born in a boston family.
He discovered that charges had two different types that where positive and negative.

Credit: C.B. Fuentes

- Purely phenomenological descriptions
- No attempts at linking with other physics

A brief history of electromagnetism

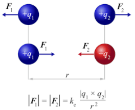
Observing nature

1780

LUIGI GALVANI (1737-1790)
Luigi was a pioneer in the field of electrophysiology, who was born at september, 9 1737 in Bologna, Italy who discovered that electricity from two different metals causes frog legs to twitch.

1790

ALESSANDRO VOLTA (1745-1827)
Born in february 18, 1745, Lombardy, Italy, Volta was an italian physicist who invented the voltaic pile battery knowing that chemistry acting on two dissimilar metals generated electricity.



1785

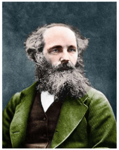
CHARLES-AUGUSTIN DE COULOMB (1736-1806)
Born june 14, 1736, in Angouleme, France, De Coulomb was a french physicist known for the formulation of Coulomb´s law, he invented the "Torsion Balance" which allowed him to measure really small charges.

1820

HANS CHRISTIAN OERSTED (1771-1851)
Born at Rudkobing, Denmark in august 14, 1777, Hans was a danish physicist and chemist who invented the compass needle that was the first connection between electricity and magnetism.

- First mathematical descriptions of specific effects
- Demonstrations that electricity can interact with biology, chemistry and magnetism

A brief history of electromagnetism
Deeper understanding



1775-1836

ANDRE MARIE AMPERE
Ampere was born January 20, 1775, Lyon, France and she was a french physicist who founded and named the science of electrodynamics, that discovered that wires carrying current produce forces on each other and that electric current produces magnetic fields.

1860

JAMES CLERK MAXWELL (1831-1879)
James was born in June 13, 1831, at Edinburgh, Scotland. He was an scottish physicist best known for his theory of electromagnetism on mathematical basis, and also James predicted electromagnetic waves.

1791-1867

MICHAEL FARADAY
Michael was born in september 22, 1791, in Surrey, England and he was a physicist and chemist who had the idea of electric field and he studied the effect of currents on magnets and magnets inducing electric current. Also he invented the disc generetor.

1880

Nikola Tesla

He discovered that electricity can past by a wireless way using waves

- More advanced mathematical descriptions of EM, first as specific phenomena and later as a unified concept
- Demonstrations that electricity can give rise to mechanical force and act without a transmission medium

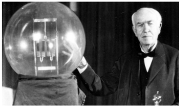
A brief history of electromagnetism
Exploitation

1885

HEINRICH HERTZ
Hertz was a brilliant german physicist and experimentalist that proved that James Clerk Maxwell was correct and he detected electromagnetic waves.

1930

THOMAS ALVA EDISON
an American inventor, scientist and businessman. He developed many devices that have had great influence around the world, such as the phonograph, the movie camera or a durable incandescent light bulb.



1895

GUGLIELMO MARCONI
Marconi was born April 25, 1874, in Bologna, Italy. He was an italian physicist and inventor of a SUCCESSFUL wireless telegraph. Also he put all his discovery to practical use by sending messages over long distances by means of radio signals. ic

- Experimental verification of Maxwell's theory
- Commercial applications

Maxwell's equations – the theory of EM

$$\vec{\nabla} \cdot \vec{E} = \frac{\rho}{\varepsilon_0}$$ Gauss' Law

$$\vec{\nabla} \cdot \vec{B} = \vec{0}$$ Gauss' Law for Magnetism

$$\vec{\nabla} \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}$$ Faraday's Law

$$\vec{\nabla} \times \vec{B} = \mu_0 \left(\vec{J} + \varepsilon_0 \frac{\partial \vec{E}}{\partial t} \right)$$ Ampere-Maxwell Law

- Originally stated as "verbal formulas"
- Mathematical formulation by Heaviside
- Dedicated experimental verification by Hertz
- First ever comprehensive *field*-based model of one of the fundamental forces in physics

Experiments and theory $\rightarrow$ applications

To name a few:

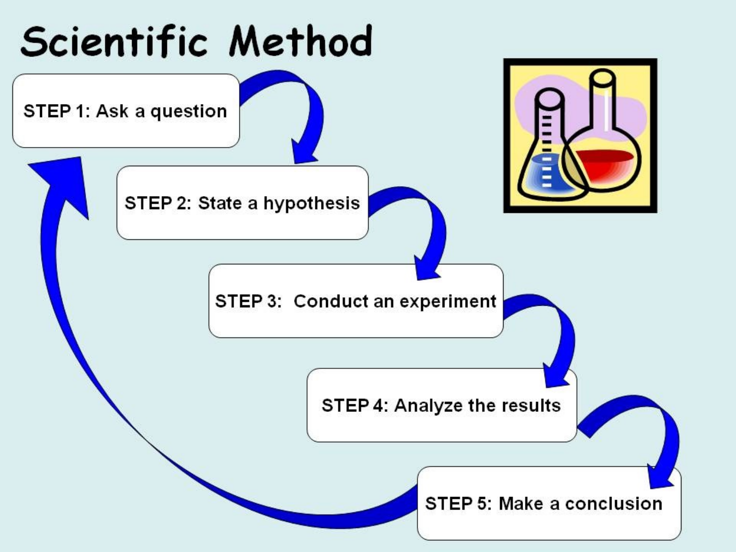
- Compass $\Rightarrow$ Age of Exploration
- Edison's light bulb $\Rightarrow$ twice as many "hours a day"
- Tesla coils $\Rightarrow$ transformer and high-voltage technology
- Telegraph $\Rightarrow$ early long-distance communication
- Turbine generators $\Rightarrow$ power grids
- Electric motors $\Rightarrow$ robots, drones
- Laser, radio, semiconductors $\Rightarrow$ computers and the Internet

Scientific inquiry

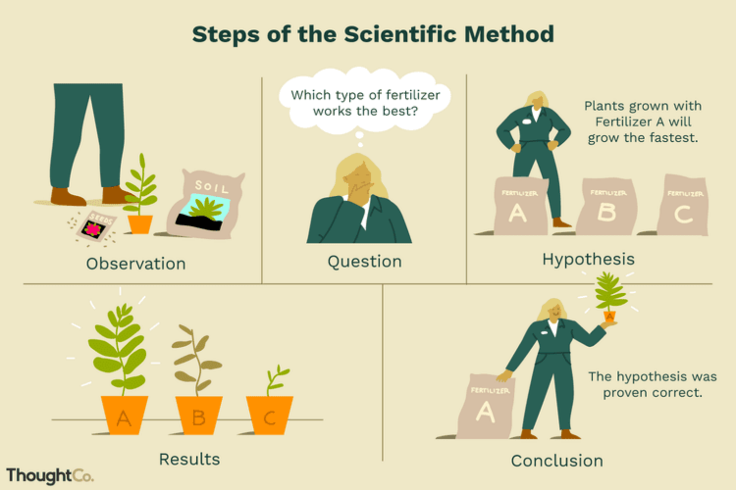
There are essentially four cornerstones in scientific inquiry:

- Characterizations (observations, definitions, and measurements of the subject of inquiry)
- Hypotheses (theoretical, hypothetical explanations of observations and measurements of the subject)
- Predictions (inductive and deductive reasoning from the hypothesis or theory)
- Experiments (tests of all of the above)

Scientific method



Scientific method
Example



Falsification

- Scientific work tends to be accepted by the scientific community when it has been confirmed. Experimental and theoretical results must be *reproduced* by others within the scientific community.
- If an experiment cannot be repeated to produce the same results, the original results might have been in error.
- Scientists need to be *skeptical* and look for ways to *falsify* proposed hypotheses.
- *Peer review* is an initial test of whether a new scientific result can be accepted.

Science $\neq$ Engineering
... most of the time

- Science is the body of knowledge that explores the physical and natural world.
- Scientists use the scientific method for the purpose of establishing new knowledge.
- Engineering is the application of knowledge in order to design, build and maintain a product or a process that solves a problem and fulfills a need.
- Engineers use scientific knowledge to create a technology.

Obviously, this division is not strict; from time to time, engineers have to act and think like scientists, and vice versa.

Outline

Why publish scientific results?

Because we must.

- The aim of science itself is uncovering “the truth”
- Scientific information is a resource
- It is in the job description
- "To work, to finish, to publish" (Faraday)

What is involved in the publication process?

Among other things,

- *Work* done in accordance with a scientific *method*
- *Writing* a publication and submitting it to an acknowledged conference, journal or similar
- *Peer review* of the work
- *Acceptance or rejection*
- ... Then either *revision or publication*

Where to submit your paper?

Two main categories of peer reviewed publications:

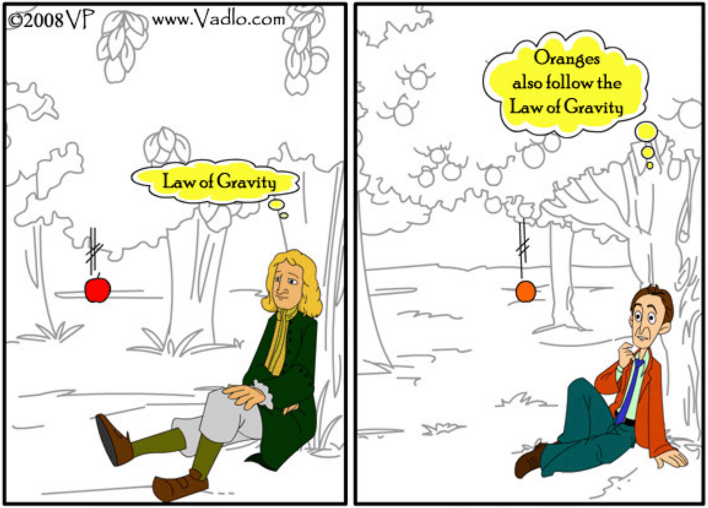
Conference proceedings – Conferences typically offer fast publication times (less than a year). Conference publications are typically short-lived and have less impact and lower rejection rates than archival journals (exceptions, e.g. computer science).

Archival Journals – Journals can have very long turn-around times (1-3 years) and often tough rejection rates. Journal publications are supposed to live 'forever' and typically have higher impacts.

Be aware of:

- Scope of publication
- Audience of publication
- Impact of publication
- For journals: is the publication listed in relevant indices?

Not all papers contain breakthroughs...



High Impact Paper　　Low Impact Paper

... but all papers *should* have a contribution!

Outline

SEMCON

SEventh seMester CONference – an exciting venue for the publication of results in most major fields of Electronics engineering:

- Audio, vision- and other media-related systems
- Control and dynamical systems
- Digital signal processing
- Networked systems
- Products and Design
- Wireless communication
- Mathematical modeling
- Robotics
- ...

Features peer review, paper revision, presentation and poster sessions.

Paper writing and review

- Go forth and do your work, then write your paper
- November 25: Lecture 2
- *Deadline for paper: November 25!*
  - You must use IEEE conference style
  - No more than 6 pages
  - If you do not have results ready: describe your *expected* results
- Beginning of December: review papers of other groups
- *Deadline for review: December 5!*
- Revise your papers according to reviews

Presentation and poster sessions

- Middle of December: Prepare slides and poster
- (Submit paper + worksheets to DE)
- December 22(probably): SEMCON presentation and poster sessions
- January: Exam - presentation, poster, scientific work method(s), reviews

Outline

Scientific work

Publication

SEMCON

Structuring a scientific paper

Summary - timeline

The ABC of science communication

The ABC of science communication is that it should be:

- **A**ccurate and **A**udience-adapted
- **B**rief
- **C**lear

Writing papers is all about communication...

- **Who** are you addressing: scientists who are specialists in your field of research, a wider group of scientists, fellow students, or public audiences?
- **Why** is your message important? Why are your communicating it? Presumably you are not doing it just for credits, but to add to the pool of knowledge, to teach, to inform, to persuade or push for development.
- **What** are your main findings or "take-home" messages? What are you going to present - new research results or a review of a topic? What prior knowledge, expectations and questions might your audience have? What technical language do they understand?
- **How** can you best deliver your message and satisfy the audience's needs? How will the audience use its new knowledge?

Getting started in writing

- Analyze your aims and your audience
  - Think of the questions: Who? Why? What? How?
- Make tables and figures of interesting results, and decide what messages to communicate
  - Think of captions that can help the figures and tables in telling their story
- Make an outline - your own style
  - should be a *working* outline
  - would normally be revised throughout the writing process
- Structure the text (here: IMRaD)
- Revise and edit

Day: The IMRaD structure

Whereas popular papers usually have a free structure, a majority of scientific papers are structured along the following lines:

- Title (6-20 words)
- Abstract (100-250 words) (SEMCON extended: 400)
- Introduction (use *Why and What For (Four)* structure)
- Methods and Materials (what to do)
- Results (don't mix methods and results)
- Discussion (conclusions in non-nature observing science)

In different scientific communities, numerous variations of the IMRaD structure coexist. It is prudent to study how other authors have structured papers in your area.

Purposes of main paper sections

Section	Intends to tell the reader
Title	What the paper is about
Abstract	Short summary that can “stand alone”
Introduction	The problem, what is known, what is not known, and the objective
Materials & Methods	What you did
Results	What you found
Discussion	How you interpret the results
Conclusions	Possible implications and the impact
Acknowledgements	Who contributed to the work and how
References	How to find the papers referred to
Appendices	Supplementary material

Papadakis: Why and what for (four)

The introduction of a paper should provide answers to the following questions:

WHY	is the topic of interest?
WHAT (1)	is the background on the previous solutions, if any?
WHAT (2)	is the background on potential solutions?
WHAT (3)	was attempted in the present effort?
WHAT (4)	will be presented in this paper?

Gives you the benefit of clarity, conciseness, simplicity, and completeness.

Introduction – Why

"The transport sector is facing a transition in the near future. As of the time of this writing, all major car manufacturers are actively engaged in designing, developing and testing autonomous vehicles, with varying success [1]."

"The global demand for poultry meat is predicted to increase by 18% between 2017 and 2027, reaching 139 billion kg [1], of which broiler meat represents the majority. Improving growth conditions in livestock stables can increase the speed at which the broilers grow without increasing feed intake, leading to significant savings worldwide."

"Low-flying drones represent an increasing potential threat to people and animals in urban areas. This paper investigates the impact of typical drones striking the human body as a result of uncontrolled descent."

Introduction – State of the Art

- Provides overview and references to previous work
- Shows what research efforts have already been undertaken
- May also outline difficulties in previous work
- Helps the reader find good references
- Shows that you (the author) know the field
- Helps setting the stage – or providing a metric – for the present work
- Helps highlighting the contribution presented in the paper

Introduction – Contribution

- Hypothesis!
  - Similar to "Final Problem Formulation", but much more narrow
  - Must be testable
  - May not always be written explicitly in the Introduction, but should be apparent to the reader
- Contribution
  - Based on everything described before, what does this paper provide of novelty?
- Outline of the rest of the paper

Scientific hypotheses

- A hypothesis is a statement that can be tested by scientific research.
    - *Dayly apple consumption leads to fewer doctor's visits*
- In a sense it is a *tentative answer* to a research question.
- A hypothesis is not just a guess—it should be based on existing theories and knowledge. It also has to be *testable*, which means you can support or refute it through scientific research methods.

Developing a scientific hypothesis

1. Ask a research question.
    - *Do students who attend more lectures get better exam results?*
2. Do some preliminary research.
    - Establish insight into the problem domain
    - Identify key variables that measure what you wish to examine
3. Make an initial formulation (one sentence).
    - *Attending more lectures leads to better exam results.*

Developing a scientific hypothesis (advanced)

4. Identify the expected outcome of an experiment to make sure the hypothesis is testable and that you can observe all relevant variables.
5. Rephrase your hypothesis to highlight the predicted relationship between key variables.
   - *The number of lectures attended by first-year students has a positive effect on their exam scores.*
6. If your research involves statistical hypothesis testing, you will also have to write a null hypothesis. The null hypothesis is the default position that there is no association between the variables.
   - H_0: *The number of lectures attended by first-year students has no effect on their final exam scores.*
   - H_1: The number of lectures attended by first-year students has a positive effect on their final exam scores.

Methods and Materials

- Assumptions and prior art
- Theory, models, basic equations
- Block diagrams
- Descriptions of data collection methods
- Algorithms
- Do NOT mix methods and results!
- To the extent possible, avoid referring to specific hardware, quantities etc.

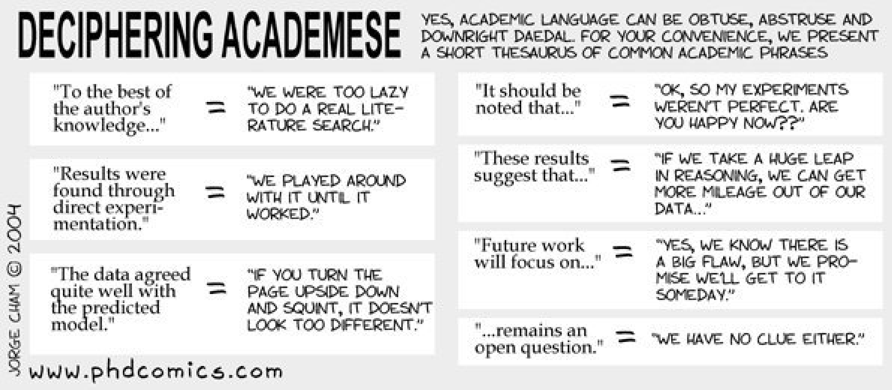
Results

- The Results section should contain
  - Detailed assumptions
  - Usage of theory and derived results
  - Presentation of specific laboratory equipment etc.
  - Descriptions of data collection results
  - Implementations/simulations
  - Data! Preferably in graphical format
- Furthermore, aim to
  - Evaluate the validity and robustness of your results
  - Help the reader understand your results.

Conclusion

- Short summary of contribution
- Key results and findings
- Interpretation of results and findings
- Future work
- If relevant: Acknowledgments

Some common academic phrases...



Outline

Summary

Publication procedure:

1. Scientific work according to proper methods
2. Write about your results in paper
3. Submit paper to a relevant conference (or journal)
4. Receive reviews and revise your papers accordingly
5. Re-submit revised paper
6. Publication

Timeline

1. Do your scientific work
2. Write a paper on it by November 25
3. Read your fellow students' papers and write reviews by December 5
4. Receive the reviews and revise your papers accordingly
5. SEMCON December 22
6. Exam in January - poster, paper AND reviews!